

Practice Problems #6

1. Let $a_1 = (1, -1, -2)$, $a_2 = (5, -4, -7)$, and $a_3 = (-3, 1, 0)$. Find the value(s) of h so that $y = (-4, 3, h)$ is a linear combination of the vectors a_1, a_2 and a_3 .
2. Is the rightmost column of the following matrix a linear combination of the first 4 columns of the matrix? If so, find all possible weights. If not, state why.

$$\begin{bmatrix} 1 & -7 & 0 & 6 & 5 \\ 0 & 0 & 1 & -2 & -3 \\ -1 & 7 & -4 & 2 & 7 \end{bmatrix}.$$

Answers on the next page

#1: $h = 5$

#2: $\{(x_1, x_2, x_3, x_4) = (5 + 7s - 6t, s, -3 + 2t, t) : s, t \in \mathbf{R}\}$